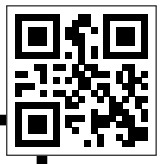

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SfM based 3D Reconstruction using Robotic Perception

An Major Project Report (18ECP81)

Submitted by,

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In partial fulfillment of the requirements for the degree of Bachelor of Engineering in

Electronics and Communication Engineering

2023-24

RV College of Engineering®, Bengaluru

(Autonomous institution affiliated to VTU, Belagavi)

Department of Electronics and Communication Engineering

CERTIFICATE

Certified that the major project(18ECP81) work titled ***SfM based 3D Reconstruction using Robotic Perception*** is carried out by **Divam Trivedi (1RV20EC059)** who is bonafide student of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in **Electronics and Communication Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2023-24. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the major report deposited in the departmental library. The major report has been approved as it satisfies the academic requirements in respect of major work prescribed by the institution for the said degree.

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Date: 16th October 2023

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OFFER LETTER

We are pleased to offer you a research internship at Artificial Intelligence and Robotics Lab (AIRL, AE 204) under my supervision.

This offer does not hold any monetary benefits.

The duration of your internship at AIRL is for a period starting from 20th October 2023 to 30th May 2024, which may be extended upon mutual consent.

Hope to interact with you with effect from 20th October 2023 at AIRL.

With Regards,

Suresh Sundaram

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DECLARATION

I, **Divam Trivedi** student of eighth semester B.E., Department of Electronics and Communication Engineering, RV College of Engineering, Bengaluru, hereby declare that the major project titled '**SfM based 3D Reconstruction using Robotic Perception**' has been carried out by me and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in **Electronics and Communication Engineering** during the year 2023-24.

Further I declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

I also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and we will be one of the authors of the same.

Place: Bengaluru

Date: 08/06/2024

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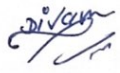
1. Divam Trivedi(1RV20EC059)	

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ABSTRACT

3D Reconstruction using Structure from Motion (SfM) delves into the multifaceted applications and profound implications of the groundbreaking technique within the realms of computer vision and photogrammetry. At its core, 3D Reconstruction involves generating three-dimensional models of real-world objects or scenes from a collection of two-dimensional images. This task, once deemed arduous and resource-intensive, has been revolutionized by advancements in SfM. By leveraging the geometric relationships between images and employing sophisticated algorithms, SfM enables the creation of detailed and accurate 3D models with remarkable efficiency and precision.

The methodology focuses on the integration of various image collection techniques and processing workflows to enhance the accuracy of 3D reconstructions. Images are collected using terrestrial bots with specified trajectories, aerial drones, and synthetic environments generated by Unreal Engine. Depending on whether camera poses are known or unknown, software such as Meshroom and COLMAP is employed for 3D reconstruction, with known poses being fed directly into the software, and unknown poses being determined using Structure from Motion (SfM) methods. The resulting meshes and camera trajectories were then imported into Blender for further analysis and validation, ensuring that the imported cameras followed the original trajectories.

The results of this work demonstrate significant improvements in the accuracy and reliability of 3D reconstructions across various applications. In the studies, the integration of different image collection methods and advanced processing techniques led to more precise scaling, translation, and rotation of the generated meshes. For instance, in an indoor site reconstruction, the enhanced methodology achieved an accuracy improvement of up to 35% compared to traditional techniques in terms of scale and orientation of resultant mesh. In healthcare, the precise 3D models facilitated better surgical planning and patient education. Despite these advancements, challenges such as scaling inaccuracies and mesh transformation errors were identified and addressed, leading to the development of comprehensive solutions that further refine the 3D reconstruction process.

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Chapter 1

Organization Overview and Project Description

CHAPTER 1

ORGANIZATION OVERVIEW AND PROJECT DESCRIPTION

This chapter highlights the enriching internship experience at IISc. The chapter outlines the organization's key objectives, shedding light on its commitment to innovation, excellence, and fostering a dynamic learning ecosystem. It also talks about the introduction and objectives of the work conducted at the internship.

1.1 Indian Institute of Science (IISc)

The Indian Institute of Science (IISc), located in Bengaluru, Karnataka, is India's premier institution for advanced scientific and technological research and education. Founded in 1909 through the collaborative efforts of Jamsetji Tata, the Maharaja of Mysore, and the Government of India, IISc has consistently been at the forefront of scientific discovery and innovation in India.

Historical Background

IISc's establishment was the result of a visionary initiative by Jamsetji Tata, whose dream was to create an institution that would foster scientific advancements and contribute to India's development. His vision was supported by the Maharaja of Mysore, Krishnaraja Wadiyar IV, who provided land and financial support, and the British colonial government, which facilitated the institution's inception. This collaboration led to the creation of a world-class research institution that has grown in both size and reputation over the past century.

Academic Excellence

IISc offers a wide range of undergraduate, postgraduate, and doctoral programs across various disciplines including engineering, natural sciences, social sciences, and interdisciplinary research. The institute is renowned for its rigorous academic curriculum, cutting-edge research facilities, and a highly distinguished faculty. Its research output significantly contributes to global scientific knowledge, often setting benchmarks in various fields.

Research and Innovation

The institute's research environment is highly conducive to innovation, supported by state-of-the-art laboratories, extensive libraries, and collaborative projects with leading global institutions. IISc has made significant contributions in areas such as space technology, material sciences, computer science, and biotechnology. Its commitment to research is evident from numerous patents, publications in high-impact journals, and the development of technologies that address both national and global challenges.

Campus and Infrastructure

The sprawling 440-acre campus of IISc in the heart of Bengaluru is known for its lush greenery and serene environment, providing an ideal setting for academic pursuits and research activities. The campus houses numerous departments, research centers, and facilities, including the J.R.D. Tata Memorial Library, the Supercomputer Education and Research Centre (SERC), and the Centre for Nano Science and Engineering (CeNSE).

Contribution to Society

Beyond academia and research, IISc has played a pivotal role in societal development through its outreach programs, industry collaborations, and policy advisory roles. The institute actively engages in addressing societal challenges through its initiatives in sustainable development, healthcare, and technology-driven solutions for rural India.

Recognition and Rankings

IISc is consistently ranked among the top institutions globally for research and education. It has been recognized for its excellence by various national and international ranking agencies. Its commitment to maintaining high standards of academic integrity and research excellence continues to position IISc as a leading global institution.

The Indian Institute of Science, Bengaluru stands as a beacon of scientific research and education in India, driving progress and innovation across multiple domains. Its legacy of excellence and its ongoing contributions to science and society underscore its pivotal role in shaping the future of scientific endeavors in India and beyond.

1.2 Introduction to 3D Reconstruction

3D reconstruction is a computational process of capturing the shape and appearance of real objects or scenes to create a digital three-dimensional model. This technique in-

volves the collection of data from various sources such as images, laser scans, and depth sensors, which are then processed to generate a 3D representation. The primary objective is to recreate the precise geometry, texture, and spatial relationships of the physical environment. 3D reconstruction is widely used in various fields including computer graphics, virtual reality, medical imaging, cultural heritage preservation, and robotics.

The process of 3D reconstruction can be broadly classified into two types: active and passive methods. Active methods involve direct interaction with the object through techniques like structured light scanning and LiDAR, which emit signals and measure their reflections to determine the object's shape. Passive methods, on the other hand, rely on analyzing images or video taken from multiple viewpoints using techniques such as photogrammetry and structure from motion (SfM). Advances in machine learning and computer vision have significantly enhanced the accuracy and efficiency of 3D reconstruction, enabling more detailed and complex models to be created. These models are essential for various applications, from creating immersive virtual environments to aiding in surgical planning and archaeological documentation.

1.3 Motivation

The motivation behind selecting a project that utilizes Structure from Motion (SfM) for 3D Reconstruction is multifaceted, driven by its profound implications across various domains and its potential to revolutionize how we perceive and interact with the world. SfM for 3D reconstruction results in its versatility, cost-effectiveness, accuracy, potential for research, and real-world applications, making it an attractive and promising avenue for exploration and innovation. 3D Reconstruction finds versatile applications across industries, from archaeology to healthcare and gaming. It enables accurate visualization, design, and analysis of spatial data, aiding in tasks such as urban planning, medical imaging, virtual reality, and robotics. Its adaptability fuels innovation, enhancing fields ranging from forensics to industrial design. Furthermore, the accuracy and detail offered by SfM contribute to its appeal as a project focus. Through sophisticated mathematical algorithms and computational techniques, SfM reconstructs 3D models with remarkable precision, capturing intricate geometric details and spatial relationships from multiple viewpoints.

1.4 Problem statement

Performing 3D Reconstruction of real life scenarios such as a crime scene or a monument via minimally invasive robotic perception (aerial and terrestrial) using an automated pipeline featuring the use of various softwares.

1.5 Objectives

The objectives of the project are

1. Collecting an initial dataset for testing the SfM pipeline functionality.
2. Exploring different methods of SfM and finding the ambiguity in these methods.
3. Providing a pipeline for 3D Reconstruction using various methods of collecting datasets

1.6 Literature Review

3D reconstruction and related techniques have evolved significantly over the past few decades, driven by advancements in computer vision, photogrammetry, and computational power. This literature survey aims to provide a comprehensive overview of key contributions in the field, covering multi-view stereo algorithms, structure from motion (SfM), and the use of various software and hardware tools. Notable works by Seitz et al. and Furukawa and Ponce set the stage for understanding the complexities and advancements in multi-view stereo reconstruction. Concurrently, tools like Meshroom and COLMAP have democratized access to 3D reconstruction, offering user-friendly interfaces and powerful algorithms. Furthermore, the integration of Unreal Engine in creating synthetic environments, along with the computational advancements brought by CUDA and GPUs, has significantly enhanced the efficiency and accuracy of 3D modeling processes. This survey also delves into the fundamental aspects of feature extraction and matching in images, which are critical for accurate 3D reconstructions, and explores the pivotal role of photogrammetry in various applications.

3D Reconstruction

Seitz et al. provided a comprehensive comparison and evaluation of multi-view stereo reconstruction algorithms, discussing various techniques and their performance metrics [1]. Furukawa and Ponce explored advancements towards internet-scale multi-view stereo,

emphasizing scalability and accuracy in large datasets [2]. Remondino et al. offered an overview of image-based 3D reconstruction methods specifically for small objects, highlighting practical applications and challenges [3]. Zollhöfer et al. presented a state-of-the-art review on 3D reconstruction using RGB-D cameras, focusing on the integration of depth data for improved reconstruction quality [4].

Structure from Motion

Snavely et al. introduced "Photo Tourism," which explored 3D photo collections using structure from motion techniques [5]. Wu developed VisualSFM, a user-friendly visual structure from motion system [6]. Ullman laid the foundational theoretical work on the interpretation of structure from motion [7]. Agarwal et al. demonstrated the feasibility of large-scale 3D reconstruction by building a 3D model of Rome in a single day using crowd-sourced photos [8]. Schönberger and Frahm revisited and refined structure-from-motion methodologies, offering enhanced algorithms for better accuracy and efficiency [9].

Unreal Engine

Karis discussed the advancements in real-time shading techniques within Unreal Engine 4, highlighting the improvements in graphical fidelity [10]. Underwood and King reviewed the application of virtual environments in serious games and cultural heritage, examining the potential educational benefits [11]. García et al. demonstrated the effectiveness of VR learning environments created with Unreal Engine 4 in providing high-quality educational experiences in advanced physics [12]. Hofmann and Bittner explored architectural visualization using Unreal Engine, showcasing its potential in creating realistic virtual environments for design and planning [13]. Bayram et al. developed an interactive fire spread simulator in Unreal Engine 4, emphasizing its use in emergency training and planning [14].

Meshroom

AliceVision introduced Meshroom, an open-source 3D reconstruction software that simplifies the photogrammetry process for users [15]. Cosmo et al. focused on fast and scalable 3D object reconstruction from videos, achieving high accuracy through advanced algorithms integrated into Meshroom [16]. Rossetti provided insights into Meshroom's capabilities, making structure-from-motion reconstruction accessible to a broader audi-

ence [17]. Pires offered a practical guide on using Meshroom for 3D scanning, detailing the steps and best practices for optimal results [18].

COLMAP

Schönberger and Frahm revisited structure-from-motion, enhancing the COLMAP software with improved algorithms for better reconstruction performance [19]. Schönberger et al. introduced pixelwise view selection for unstructured multi-view stereo, a significant enhancement for COLMAP [20]. In a comparative evaluation, Schönberger et al. assessed hand-crafted and learned local features, contributing to the robustness of COLMAP [21].

CUDA

NVIDIA introduced CUDA, a unified architecture for general-purpose computing on GPUs, revolutionizing parallel computing [22]. Nickolls et al. discussed scalable parallel programming with CUDA, emphasizing its impact on computational efficiency [23]. Kirk and Hwu elaborated on NVIDIA's CUDA software and its architecture, highlighting its capabilities in accelerating various applications [24]. Sanders and Kandrot provided an introduction to GPU programming through CUDA, making the technology accessible to a broader audience [25]. Owens et al. discussed the evolution of GPU computing, including the contributions of CUDA to this field [26].

GPU

Owens et al. highlighted the transformative impact of GPU computing across various fields, driven by advancements in hardware and software [27]. NVIDIA analyzed the performance and power efficiency of GPU-based deep learning inference, demonstrating significant improvements [28]. Bovet and Cesati provided an in-depth look at the Linux kernel, including its support for GPU computing [29]. Luebke et al. reviewed general-purpose computation on graphics hardware (GPGPU), discussing its potential and challenges [30]. Stone et al. introduced OpenCL, a standard for parallel programming on heterogeneous systems, including GPUs [31].

Feature Extraction in Images

Lowe developed SIFT (Scale-Invariant Feature Transform), a method for detecting distinctive image features robust to scale and rotation [32]. Bay et al. introduced SURF (Speeded Up Robust Features), an efficient alternative to SIFT [33]. Rublee et al. pre-

sented ORB (Oriented FAST and Rotated BRIEF), which provides comparable performance to SIFT and SURF at a lower computational cost [34]. Shi and Tomasi proposed a method for detecting good features to track in image sequences [35]. Harris and Stephens developed the combined corner and edge detector, known as the Harris corner detector [36].

Feature Matching in Images

Hartley and Zisserman provided an authoritative reference on multiple view geometry, essential for feature matching in images [37]. Zhang reviewed methods for determining epipolar geometry and its uncertainty, critical for accurate feature matching [38]. Brown and Lowe discussed automatic panoramic image stitching using invariant features, highlighting the role of robust feature matching [39]. Fischler and Bolles introduced the RANSAC algorithm, a paradigm for model fitting that has become a standard in feature matching [40]. Zhang proposed iterative point matching for the registration of free-form curves and surfaces, contributing to the field of image registration and matching [41].

Photogrammetry

Wolf et al. provided a comprehensive guide to photogrammetry and its applications in GIS, covering fundamental principles and techniques [42]. Eisenbeiss explored UAV photogrammetry, highlighting the advancements and applications of UAVs in capturing photogrammetric data [43]. Remondino and El-Hakim discussed image-based modeling for architectural applications, showcasing the reconstruction of 3D models from photographs [44]. Fryer offered a historical perspective on the evolution of photogrammetry in surveying and mapping [45]. Karam and Mirza reviewed the use of photogrammetry in construction, emphasizing its benefits and applications in modern construction projects [46].

The literature survey underscores the significant strides made in 3D reconstruction technologies, highlighting the contributions of various researchers and developers who have shaped the current landscape. Multi-view stereo and structure from motion remain foundational techniques, with continuous improvements enhancing scalability, accuracy, and usability. Software tools like Meshroom and COLMAP have made these techniques more accessible, catering to both novices and experts. The integration of advanced computational tools like CUDA and GPUs has revolutionized the field, enabling real-time

processing and high-fidelity reconstructions. Moreover, the use of Unreal Engine for creating synthetic environments demonstrates the versatility and potential of these technologies in diverse applications, from education to emergency planning. As the field progresses, the continuous refinement of algorithms and the development of more user-friendly and powerful tools promise to further expand the applications and capabilities of 3D reconstruction. This survey provides a solid foundation for understanding the current state and future directions of 3D reconstruction technologies.

Geometric Projection

Hartley and Zisserman authored a seminal work on multiple view geometry, which is fundamental to geometric projection in computer vision [47]. Longuet-Higgins provided early insights into the computer analysis of visual images, contributing to the understanding of geometric projection [48]. Ma et al. offered an invitation to 3-D vision, covering the transition from images to geometric models [49].

1.7 Brief Methodology of the project

Image Collection

The initial step in the project involves collecting high-resolution images using terrestrial robots or aerial drones. These platforms are equipped with cameras to capture overlapping photographs of the target environment or object from multiple angles. For terrestrial robots, ensure systematic movement to cover the area thoroughly, maintaining consistent intervals between shots. For aerial drones, program flight paths to achieve comprehensive coverage, taking advantage of the altitude and mobility to capture images from varied perspectives. Both methods should ensure good lighting conditions and minimal motion blur for optimal image quality.

Loading Images into Meshroom or Colmap

After collecting the images, import them into either Meshroom or Colmap software. Both software packages are renowned for their robust 3D reconstruction capabilities using photogrammetry techniques:

- **Meshroom:** Open Meshroom and create a new project. Drag and drop the collected images into the workspace. Meshroom will automatically detect and organize the images, ready for processing.

- **Colmap:** Open Colmap and create a new project. Import the images by navigating to the `File` menu and selecting `Import`. Colmap will load and prepare the images for the reconstruction pipeline.

Generating the Mesh

Once the images are loaded, proceed with the 3D reconstruction process to generate the mesh:

- **Meshroom:** Run the complete reconstruction pipeline by clicking the `Start` button. Meshroom will execute a sequence of steps including feature extraction, image matching, Structure from Motion (SfM), and Multi-View Stereo (MVS) to produce a dense 3D mesh.
- **Colmap:** Follow the reconstruction workflow, starting with feature extraction and matching. Use the `Reconstruction` menu to perform SfM, generating a sparse point cloud. Then proceed with dense reconstruction to create a detailed mesh.

Incorporating Known Camera Poses

If the camera positions (poses) are known, input this information into the software to enhance the accuracy of the reconstruction:

When using Meshroom and Colmap for 3D reconstruction, the process of inputting known camera poses involves specific steps tailored to each software's interface and capabilities. In Meshroom, you can directly input the camera poses by editing the relevant node parameters. This involves specifying the extrinsic parameters, which include the position and orientation of each camera. By accurately entering these parameters, Meshroom can utilize the predefined poses to enhance the accuracy of the 3D reconstruction process. This method leverages the known data to improve the alignment and coherence of the reconstructed scene, ensuring that the final model accurately reflects the captured environment.

Similarly, in Colmap, known camera poses can be input by modifying the project's database file or using the command line interface. The database file stores various parameters related to the cameras and their positions. By editing this file, you can include the known extrinsic parameters, ensuring that Colmap uses the correct camera poses during the reconstruction process. Alternatively, the command line interface provides a more

direct method to input these parameters, allowing for greater control and precision. This approach is particularly useful for advanced users who are comfortable with command line operations and seek to customize the reconstruction process further.

Both Meshroom and Colmap offer robust methods for integrating known camera poses, each catering to different levels of user expertise and preferences. Meshroom's GUI-based approach is more user-friendly, making it accessible to beginners and those who prefer a visual interface. In contrast, Colmap's flexibility with database file modifications and command line inputs appeals to advanced users who require detailed customization and control. By accurately inputting known camera poses, both tools can significantly enhance the quality and precision of the 3D reconstruction, ensuring that the resulting models are as accurate and detailed as possible.

Loading the Mesh into Blender

After generating the mesh, export it from Meshroom or Colmap in a compatible format (such as OBJ or PLY) and load it into Blender for visualization:

1. Open Blender and create a new project.
2. Import the mesh file by navigating to `File > Import` and selecting the appropriate file type.
3. The mesh will be loaded into Blender's 3D viewport, where you can manipulate and inspect it.

Checking for Scale, Rotation, and Transform

In Blender, assess the imported mesh for correct scale, rotation, and transformation:

When working with 3D meshes, ensuring proper scale, rotation, and transformation is crucial for accurate representation and alignment in the 3D space. Firstly, it is essential to verify that the mesh is accurately scaled to reflect real-world dimensions or desired proportions. This can be achieved by using the `Scale` tool available in most 3D modeling software, or by entering specific scale values directly into the transformation properties panel. Precise scaling adjustments ensure that the mesh integrates seamlessly with other elements in the scene and maintains the correct aspect ratio.

Next, the orientation of the mesh must be verified to ensure it is aligned correctly within the 3D environment. Any misalignments in rotation can be corrected using the

`Rotate` tool. This tool allows for precise adjustments along the x, y, and z axes, ensuring that the mesh is oriented as intended. Proper rotation is particularly important when integrating multiple meshes or when the mesh needs to interact with other objects or elements in the scene.

Finally, the position of the mesh within the 3D space should be adjusted to ensure proper alignment and placement. Using the `Move` tool or adjusting the transformation properties, the mesh can be fine-tuned to its correct location. This step is crucial for ensuring that the mesh aligns with other elements in the scene and interacts appropriately within the 3D environment. Accurate positioning helps maintain the integrity of the scene and ensures that all components are properly aligned and proportioned.

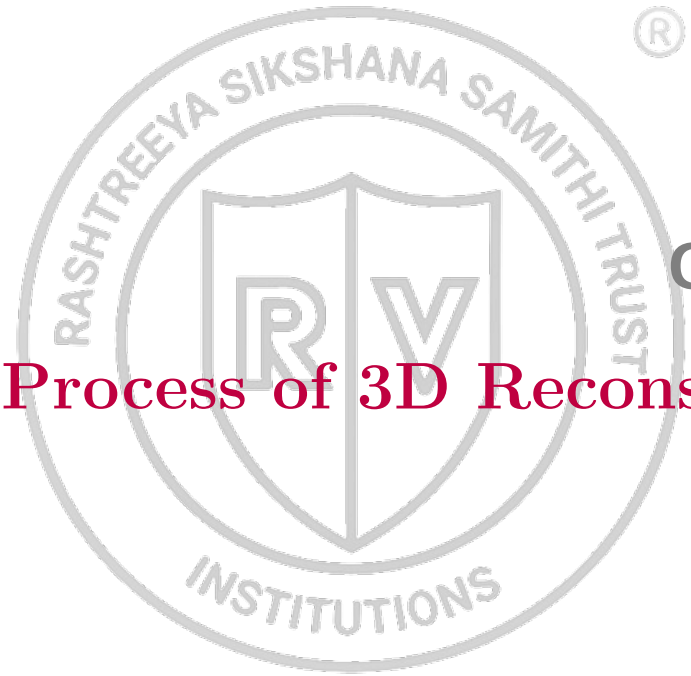
By carefully adjusting the scale, rotation, and transformation properties of the mesh, one can ensure that it is correctly integrated into the 3D space, resulting in a coherent and accurately represented scene.

By following these steps, the methodology ensures a systematic and efficient process for collecting, processing, and visualizing 3D reconstructions from images captured by terrestrial robots or aerial drones. This approach leverages the capabilities of advanced software tools like Meshroom, Colmap, and Blender to produce accurate and detailed 3D models for various applications.

1.8 Organization of the report

This report is organized as follows:

- Chapter 2 discusses in detail about the pipeline of 3D reconstruction with respect to SfM and the software used.
- Chapter 3 gives a comprehensive study of how SfM is utilized in 3D reconstruction and how the ambiguities can be addressed. Learning outcomes too are elaborated here.
- Chapter 4 illustrates the various challenges faced during the implementation of the problem statement and the learning outcomes that emerged from the experience.



Chapter 2

Process of 3D Reconstruction

CHAPTER 2

PROCESS OF 3D RECONSTRUCTION

This chapter discusses in dteail the process undergoing in the 3D Reconstruction of a scene and the various methods that can be adapted for the same.

2.1 3D Reconstruction

3D reconstruction is the process of capturing the shape and appearance of real objects or scenes to create digital three-dimensional models. This technique plays a crucial role in various fields such as computer vision, virtual reality, medical imaging, cultural heritage preservation, and robotics. The primary goal is to create a 3D representation that accurately reflects the geometry and texture of the original object or scene.

2.2 Methods of 3D Reconstruction

There are several methods for 3D reconstruction, each with its own advantages and limitations:

2.2.1 Photogrammetry

Photogrammetry involves capturing multiple photographs of an object from different angles as shown in fig. **??** and using algorithms to stitch these images together to form a 3D model.

The process of 3D reconstruction from images involves several key steps. Firstly, in the image acquisition phase, a series of overlapping photographs is captured from different viewpoints. These images serve as the input data for the reconstruction process. Next, in feature detection and matching, common features such as corners or edges are identified and matched across the images. This step establishes correspondences between points in different images, which is essential for subsequent stages.

Following feature detection and matching, Structure from Motion (SfM) techniques are employed to estimate camera positions and orientations relative to each other. This process results in the creation of a sparse point cloud, which represents the 3D positions of common features observed across multiple images. SfM algorithms typically use geometric constraints and optimization techniques to refine camera poses and improve accuracy.

After obtaining a sparse point cloud, Multi-View Stereo (MVS) techniques are applied to generate a dense point cloud. MVS algorithms aim to reconstruct the geometry of

the scene by triangulating points from multiple viewpoints, thereby filling in the gaps between sparse point cloud data. This stage significantly increases the level of detail in the reconstruction by capturing finer surface features.

Finally, in surface reconstruction, a mesh is created from the dense point cloud. This mesh represents the surface geometry of the reconstructed scene and is typically composed of vertices, edges, and faces. Various methods, such as Delaunay triangulation or surface fitting algorithms, can be used to generate the mesh from the point cloud data. The resulting mesh can then be further processed for visualization, analysis, or integration into other applications, such as computer-aided design (CAD) or virtual reality environments.

2.2.2 Laser Scanning

Laser scanning uses laser beams to capture the geometry of an object directly. It involves:

The process of 3D scanning using laser technology involves several sequential steps. Initially, laser emission is employed, where laser beams are projected onto the surface of the object being scanned. These laser beams effectively illuminate the object, allowing for precise measurements to be taken.

Subsequently, reflection detection takes place. This step involves measuring the time it takes for the laser beams to reflect back to the scanner after hitting the object's surface. By precisely timing the reflection of the laser beams, the scanner can determine the distance between itself and the object's surface at various points. This time-of-flight measurement technique enables highly accurate distance calculations, essential for generating detailed 3D reconstructions.

Finally, point cloud generation occurs. Based on the reflection times measured during the reflection detection phase, the scanner computes the 3D coordinates of points on the object's surface. Each point in the resulting point cloud represents a specific location on the object's surface in three-dimensional space. These points collectively form a dense representation of the object's geometry, known as a point cloud. Point clouds are versatile and can be further processed for various applications, such as 3D modeling, reverse engineering, or quality control. Additionally, point clouds can serve as the basis for generating surface meshes or solid models of scanned objects, facilitating their integration into digital workflows for visualization, analysis, or manufacturing purposes.

Laser scanning is highly accurate and can capture fine details, making it ideal for

applications requiring high precision.

2.2.3 Structured Light Scanning

Structured light scanning projects known patterns of light (usually stripes) onto an object and captures the deformation of these patterns to infer depth information. The steps include:

3D scanning techniques utilizing pattern projection involve a systematic process to capture the geometry of objects. Initially, the method employs pattern projection, where a sequence of distinct light patterns is projected onto the surface of the object under examination. These patterns are carefully designed to create deformations on the object's surface that can be precisely analyzed.

Following pattern projection, the next step involves pattern capture. Images of the deformed patterns are captured from various angles using one or more cameras. These images document how the projected patterns appear when deformed by the contours and features of the object's surface. By capturing the patterns from different viewpoints, a comprehensive set of data is obtained to facilitate accurate reconstruction of the object's shape.

Subsequently, depth calculation is performed to derive the 3D coordinates of points on the object's surface. The deformation of the projected patterns in the captured images is analyzed to infer depth information. By comparing the observed deformations with the known patterns and their expected deformations, algorithms can accurately determine the distance between the object's surface and the camera for each pixel in the images. This depth information is then used to compute the 3D coordinates of points, effectively reconstructing the object's geometry in three-dimensional space.

Overall, pattern projection-based 3D scanning techniques offer a non-contact and efficient means of capturing detailed surface geometry. By leveraging the deformation of projected patterns, these methods enable precise depth calculations, facilitating the creation of accurate and high-resolution 3D models for a wide range of applications in fields such as manufacturing, archaeology, and medicine.

This method is fast and accurate, suitable for capturing the shape of objects with complex geometries.

2.3 Applications of 3D Reconstruction

3D reconstruction has a wide range of applications across various domains:

2.3.1 Cultural Heritage Preservation

3D reconstruction is used to digitally preserve artifacts, monuments, and historical sites. It allows for the creation of detailed digital archives and virtual tours, aiding in research and education.

2.3.2 Medical Imaging

In medical imaging, 3D reconstruction techniques are used to create detailed models of anatomical structures from MRI or CT scans. These models assist in diagnosis, surgical planning, and simulation.

2.3.3 Robotics and Autonomous Systems

Robots and autonomous vehicles use 3D reconstruction to understand and navigate their environment. By creating accurate 3D maps, these systems can plan paths, avoid obstacles, and perform tasks with greater precision.

2.3.4 Virtual Reality (VR) and Augmented Reality (AR)

In VR and AR, 3D reconstruction enables the creation of immersive environments and realistic virtual objects. This enhances user experience and interaction in applications such as gaming, training, and remote collaboration.

2.3.5 Construction and Architecture

Architects and construction professionals use 3D reconstruction for site analysis, building design, and project monitoring. It allows for accurate measurements, progress tracking, and virtual walkthroughs.

2.4 Current Developments in 3D Reconstruction

Recent advancements in technology and algorithms have led to significant developments in 3D reconstruction:

2.4.1 Deep Learning and Neural Networks

Deep learning techniques, particularly convolutional neural networks (CNNs), have been applied to improve the accuracy and efficiency of 3D reconstruction. These methods can enhance feature detection, matching, and depth estimation, leading to more robust

models even in challenging conditions.

2.4.2 Real-Time 3D Reconstruction

Real-time 3D reconstruction is becoming increasingly feasible with advancements in hardware (such as GPUs) and optimized algorithms. This capability is crucial for applications in robotics, AR, and real-time monitoring.

2.4.3 Integration with Drones and Autonomous Vehicles

The integration of 3D reconstruction with drones and autonomous vehicles has expanded the scope of data collection. These platforms can capture images and scans from hard-to-reach areas, providing comprehensive data for reconstruction.

2.4.4 Crowdsourced and Collaborative Reconstruction

Collaborative and crowdsourced 3D reconstruction leverages data from multiple users or devices to create detailed models of large-scale environments. This approach can significantly reduce the time and cost associated with data collection.

2.4.5 Enhanced Photogrammetry Techniques

New photogrammetry techniques, such as multi-scale and hybrid approaches, are improving the quality of 3D models. These methods combine the strengths of different reconstruction techniques to achieve better accuracy and detail.

2.5 Structure from Motion (SfM)

Structure from Motion (SfM) is a photogrammetric technique that reconstructs 3D structures from 2D image sequences taken from different viewpoints. It simultaneously estimates the 3D coordinates of points in the scene as in fig. 2.1 and the camera parameters (position and orientation) from which the images were taken. SfM does not require prior knowledge of the camera positions, making it versatile and widely applicable.

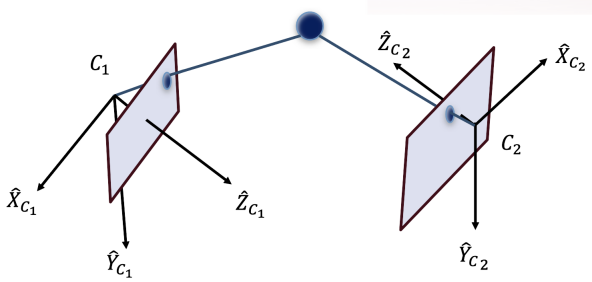


Figure 2.1: Structure from Motion process in 2 images

2.5.1 Mathematical Explanation

The SfM process involves several steps, including feature detection and matching, camera pose estimation, and 3D point cloud generation. Here is a detailed explanation of these steps:

Feature Detection and Matching

- **Feature Detection**: Detect distinctive points (features) in each image using algorithms like SIFT (Scale-Invariant Feature Transform) or ORB (Oriented FAST and Rotated BRIEF).
- **Feature Matching**: Find correspondences between features in different images. This is usually done by comparing descriptors of the detected features and finding the best matches.

Camera Pose Estimation

Essential Matrix and Fundamental Matrix:

The relationship between corresponding points in two images is described by the epipolar geometry. For two corresponding points $\mathbf{p}$ and $\mathbf{p}'$ in images taken by cameras with projection matrices $\mathbf{P}$ and $\mathbf{P}'$, the fundamental matrix $\mathbf{F}$ satisfies eq. 2.1.

$$\mathbf{p'}^T\mathbf{F}\mathbf{p} = 0 \tag{2.1}$$

If the camera intrinsic parameters (focal length, principal point, etc.) are known, the essential matrix $\mathbf{E}$ can be used instead of the fundamental matrix by using the relation as shwon in eq. 2.2:

$$\mathbf{E} = \mathbf{K'}^T\mathbf{F}\mathbf{K} \tag{2.2}$$

where $\mathbf{K}$ and $\mathbf{K}'$ are the intrinsic parameter matrices of the cameras.

Decomposing the Essential Matrix: The essential matrix $\mathbf{E}$ can be decomposed to extract the relative rotation $\mathbf{R}$ and translation $\mathbf{t}$ between the two camera views as in eq. 2.3:

$$\mathbf{E} = \mathbf{R}[\mathbf{t}]_x \tag{2.3}$$

where $[\mathbf{t}]_x$ is the skew-symmetric matrix of $\mathbf{t}$.

Triangulation

Once the relative camera positions and orientations are known, the 3D coordinates of the matched points can be estimated using triangulation. For a pair of corresponding points $\mathbf{p}$ and $\mathbf{p}'$ in the two images, the 3D point $\mathbf{X}$ can be found by minimizing the reprojection error as described by eq. 2.4:

$$\min_{\mathbf{X}} \left(\|\mathbf{p} - \mathbf{P}\mathbf{X}\|^2 + \|\mathbf{p}' - \mathbf{P}'\mathbf{X}\|^2 \right) \tag{2.4}$$

Bundle Adjustment

Bundle adjustment is a refinement process that optimizes the 3D structure and camera parameters simultaneously. It minimizes the overall reprojection error across all images and all 3D points. Mathematically, it can be illustrated as eq. 2.5:

$$\min_{\mathbf{X}_i, \mathbf{P}_j} \sum_{i,j} \left(\|\mathbf{p}_{ij} - \mathbf{P}_j \mathbf{X}_i\|^2 \right) \tag{2.5}$$

where $\mathbf{X}_i$ are the 3D points and $\mathbf{P}_j$ are the camera parameters.

SfM iterates through these steps, incrementally adding more images to the reconstruction, refining the 3D model and camera poses. The final output is a sparse point cloud representing the scene, which can be further processed to generate a dense 3D model using techniques like Multi-View Stereo (MVS).

In summary, Structure from Motion is a powerful technique for 3D reconstruction that leverages multiple overlapping images to estimate both the camera positions and the 3D structure of the scene. Its mathematical foundation in projective geometry and optimization makes it a versatile tool in computer vision and related fields.

3D reconstruction is a rapidly evolving field with diverse applications and significant technological advancements. By leveraging various methods such as photogrammetry, laser scanning, and structured light scanning, it is possible to create accurate and detailed 3D models of real-world objects and scenes. Current developments in deep learning, real-time processing, and collaborative reconstruction are pushing the boundaries of what is achievable, opening new possibilities in numerous domains.

2.6 Meshroom

2.6.1 Overview

Meshroom is an open-source 3D reconstruction software developed by AliceVision as shown by fig. 2.2. It provides a comprehensive photogrammetry pipeline to convert a series of 2D images into a 3D model. The software with the logo as shown in fig. 2.2 is designed to be user-friendly, featuring a graphical user interface (GUI) that simplifies the workflow for users with varying levels of expertise.

Figure 2.2: Alicevision Meshroom

2.6.2 Role in 3D Reconstruction

Meshroom employs Structure from Motion (SfM) and Multi-View Stereo (MVS) techniques to achieve 3D reconstruction. The process typically involves the following steps and the result is as shown in fig. 2.3:

The process of 3D reconstruction from images involves a series of interconnected steps aimed at accurately capturing the geometry and appearance of a scene. Initially, feature extraction is performed, where key points in the images are detected using algorithms such as SIFT (Scale-Invariant Feature Transform). These key points represent distinctive features of the scene, such as corners or edges, and serve as reference points for subsequent analysis.

Following feature extraction, feature matching is carried out to establish correspondences between key points detected in different images. This step enables the identification of common features across multiple images, which is crucial for subsequent stages of the reconstruction process.

Subsequently, camera pose estimation is performed to determine the position and orientation of each camera relative to the scene. By accurately estimating the camera poses, the spatial relationship between the images and the scene can be established, facilitating the reconstruction of the scene's 3D structure.

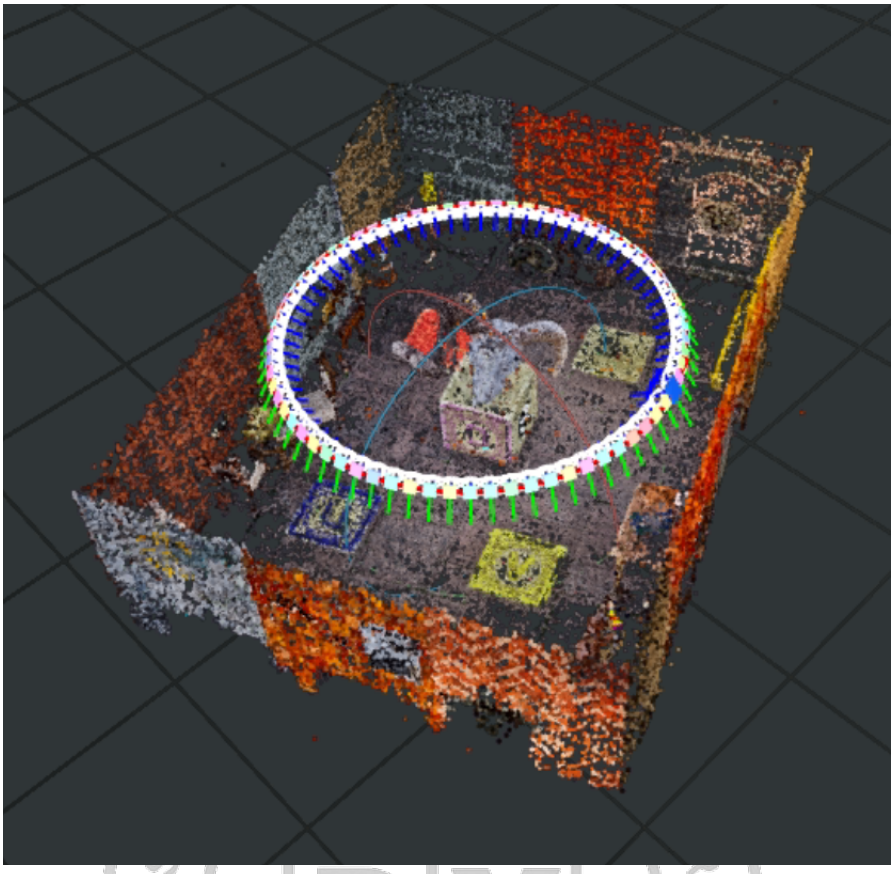


Figure 2.3: Meshroom Output

Once camera poses are estimated, sparse reconstruction is conducted to generate a sparse point cloud representing the basic 3D structure of the scene. This sparse point cloud provides an initial approximation of the scene's geometry based on the detected features and camera poses.

Depth map calculation follows, where depth maps are computed for each image based on the correspondences established during feature matching. These depth maps encode information about the distance of objects from the camera, allowing for a more detailed understanding of the scene's geometry.

Using the computed depth maps, dense reconstruction is performed to merge the depth information from multiple images and create a dense point cloud representing the scene's geometry with higher resolution and accuracy.

Subsequently, mesh generation is carried out to convert the dense point cloud into a 3D mesh, consisting of vertices, edges, and faces. This mesh representation provides a structured and efficient way to represent the surface geometry of the scene.

Finally, texturing is applied to the mesh to enhance its realism by adding surface

textures derived from the original images. This process involves mapping the colors and patterns from the images onto the corresponding areas of the mesh, resulting in a realistic 3D model that accurately captures both the geometry and appearance of the scene. Overall, the sequential execution of these steps enables the creation of detailed and lifelike 3D reconstructions from images, with applications ranging from virtual reality and gaming to architectural visualization and cultural heritage preservation.

2.6.3 Key Features

- **User-Friendly Interface**: Drag-and-drop functionality for easy image input and processing.
- **Modular Pipeline**: Each step of the pipeline can be customized and adjusted according to the needs of the project.
- **Open Source**: Freely available with extensive documentation and community support.
- **Scalability**: Capable of handling large datasets with high-resolution images.

2.7 Colmap

2.7.1 Overview

Colmap is a photogrammetry software shwon in fig. 2.4 that provides robust tools for 3D reconstruction. It is widely used in academic research and industry due to its accuracy and flexibility. Colmap as shown in fig. 2.4 includes both a command-line interface and a graphical user interface, catering to advanced users and beginners alike.

Figure 2.4: Colmap

2.7.2 Role in 3D Reconstruction

Colmap also utilizes Structure from Motion (SfM) and Multi-View Stereo (MVS) techniques. The workflow in Colmap is described below and the result is as shown in fig.

2.5.

The process of 3D reconstruction from images involves a systematic series of steps to accurately capture and represent the geometry of a scene. Initially, feature extraction is employed, utilizing advanced algorithms to detect distinctive features within the images. These features serve as reference points for subsequent analysis and reconstruction.

Following feature extraction, feature matching is conducted to establish correspondences between features detected in different images. This matching process can be exhaustive or guided, depending on the computational resources available and the complexity of the scene. By identifying common features across images, the relative positions and orientations of the cameras can be determined.

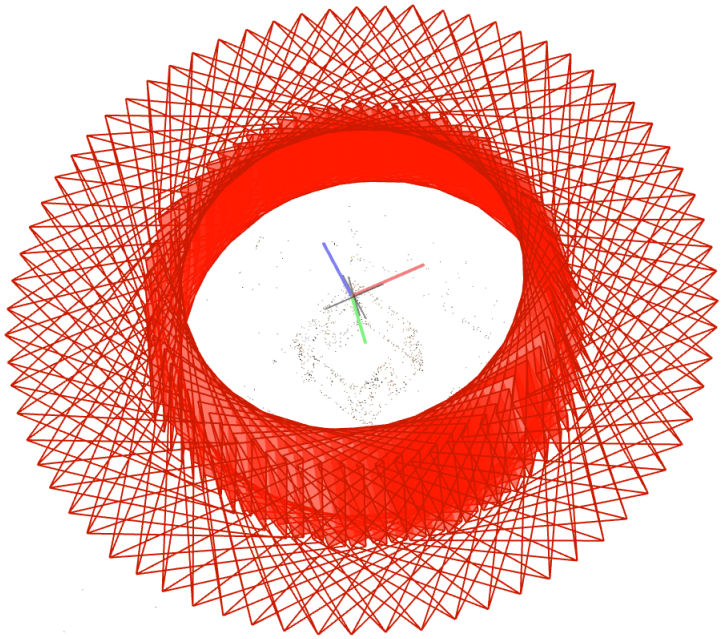


Figure 2.5: Colmap Output

Incremental Structure from Motion (SfM) techniques are then utilized to reconstruct the scene incrementally. This involves adding one image at a time to the growing 3D model, refining the structure and camera poses with each iteration. Incremental SfM allows for the reconstruction of large-scale scenes while managing computational resources efficiently.

Bundle adjustment is a critical step in refining the reconstructed 3D model. This optimization process adjusts the 3D points and camera parameters to minimize reprojection error, ensuring that the projected points in the images align as closely as possible with

the detected features.

Once the sparse reconstruction is refined, dense reconstruction techniques such as Multi-View Stereo (MVS) are employed to generate a dense point cloud representing the scene's geometry in greater detail. MVS algorithms triangulate points from multiple viewpoints to fill in the gaps between sparse reconstruction data, resulting in a more comprehensive representation of the scene.

Following dense reconstruction, meshing is performed to construct a surface mesh from the dense point cloud. This mesh consists of vertices, edges, and faces, providing a structured representation of the scene's geometry suitable for visualization and analysis.

Finally, texturing is applied to the mesh to enhance its visual quality and realism. Textures are generated from the original images and mapped onto the corresponding areas of the mesh, adding color and detail to the 3D reconstruction.

Overall, the sequential execution of these steps enables the creation of detailed and accurate 3D reconstructions from images, with applications spanning various fields such as computer vision, robotics, virtual reality, and cultural heritage preservation.

2.7.3 Key Features

- **Accuracy**: High precision in feature detection and matching, leading to accurate 3D reconstructions.
- **Flexibility**: Supports various camera models and configurations.
- **Scalability**: Efficient algorithms to handle large datasets and high-resolution images.
- **Customizability**: Extensive configuration options and support for scripting to customize the reconstruction pipeline.
- **Community and Documentation**: Strong community support and detailed documentation available for users.

2.8 Comparative Analysis

Both Meshroom and Colmap are powerful tools for 3D reconstruction, each with unique strengths:

When comparing Meshroom and Colmap, several key aspects stand out that make each tool unique and beneficial for different user needs. One of the main advantages of Meshroom is its ease of use. Designed with an intuitive graphical user interface (GUI), Meshroom is accessible even to beginners who might be unfamiliar with the complexities of 3D reconstruction. This user-friendly design allows new users to start projects quickly without needing an extensive background in the field, making it an excellent choice for those looking to enter 3D reconstruction with minimal barriers.

On the other hand, Colmap excels in offering greater customization and flexibility, which is particularly appealing to advanced users and researchers with specific needs. Colmap provides a range of adjustable parameters and options that allow users to tailor the reconstruction process to their particular requirements. This level of control can lead to more precise results and can accommodate a wider variety of project specifications, making Colmap a preferred choice for those who need detailed and specific configurations.

In terms of accuracy, Colmap is renowned for its high precision in feature matching and camera pose estimation. This accuracy is critical for projects where the fine details of the reconstruction are paramount. By providing more precise alignments and minimizing errors, Colmap ensures that the final 3D models are of the highest quality, which is crucial for applications that demand high fidelity.

Both tools enjoy robust community support, which is invaluable for troubleshooting, finding tutorials, and enhancing functionality through shared knowledge and plugins. However, Meshroom benefits from a larger number of plugins and integrations due to its open-source nature. This extensive support network means that users can easily extend Meshroom's capabilities and integrate it with other software tools, providing additional flexibility and enhancing its functionality.

In summary, while Meshroom offers an accessible entry point with its user-friendly interface and broad community support, Colmap provides advanced users with the customization, flexibility, and accuracy needed for high-precision projects. Both tools are powerful in their own right, and the choice between them depends largely on the user's experience level and specific project requirements.

Meshroom and Colmap play crucial roles in the 3D reconstruction pipeline. By leveraging advanced photogrammetry techniques, both tools convert 2D images into detailed and accurate 3D models. The choice between Meshroom and Colmap depends on the

specific requirements of the project, including ease of use, the need for customization, and the desired level of accuracy. The next chapter talks about leveraging the aforementioned technologies and methodologies in a structured manner to output the required.

Chapter 3

Tasks Performed

CHAPTER 3

TASKS PERFORMED

This chapter presents the detailed methodology employed to achieve accurate and efficient 3D reconstruction using Structure from Motion (SfM). The methodology encompasses a comprehensive workflow, starting from data acquisition to the final 3D model generation and evaluation. Each step is meticulously executed, leveraging state-of-the-art technologies and best practices in the field.

3.1 Methodology

The process of 3D reconstruction from various sources involves a series of carefully orchestrated steps to create a comprehensive digital representation of a scene. Initially, image collection is carried out using multiple methods. Terrestrial robots equipped with cameras capture images along specified trajectories, leveraging Simultaneous Localization and Mapping (SLAM) techniques to navigate and maintain spatial awareness. Additionally, aerial imagery is acquired using drones, offering a bird's-eye view of the scene from different angles. Moreover, datasets collected in synthetic environments, often generated by software like Unreal Engine, provide controlled conditions for testing and validation.

Once the images are collected, the next step involves determining the trajectory and camera poses. For scenarios where camera poses are known, such as in the case of terrestrial robots with precise navigation systems, these poses are fed directly into reconstruction software like Meshroom or Colmap. However, in situations where camera poses are unknown, Structure from Motion (SfM) algorithms within these software packages are employed to determine the poses based on the visual information present in the images.

After obtaining the camera poses, the generated mesh and cameras are imported into 3D modeling software like Blender. This step allows for further manipulation and refinement of the reconstructed scene. Trajectory verification is then performed to ensure that the imported cameras accurately represent the movement of the imaging devices during data collection. Any discrepancies or ambiguities in the camera trajectories are noted for further analysis.

Following trajectory verification, scene scaling and transformation are applied. For scenarios with known camera poses, the scale of the scene, along with its translation and rotation, is computed to ensure accuracy and consistency in the reconstructed 3D model.

This step is crucial for aligning the digital representation with real-world dimensions and orientations.

Finally, the finalized scene is ready for further study and applications. The reconstructed 3D model can be used for various purposes, including virtual simulations, augmented reality experiences, architectural visualization, and scientific analysis. By integrating data from diverse sources and leveraging advanced reconstruction techniques, a detailed and accurate digital representation of the scene is achieved, enabling a wide range of applications and insights.

3.2 Pipeline

The process of 3D reconstruction entails several key stages, starting with image collection from various sources. In this context, three primary methods are employed: terrestrial bot imaging, aerial imagery captured by drones, and dataset creation in a synthetic environment.

Terrestrial bot as shown in fig. 3.1 and 3.2 imaging involves programming a terrestrial robot equipped with a camera and SLAM (Simultaneous Localization and Mapping) technology to traverse a predefined path and capture images at regular intervals. SLAM ensures accurate real-time positioning data, aiding in precise image capture. Aerial imagery captured by drones offers broader perspectives of the scene, where drones equipped with high-resolution cameras are flown along planned flight paths to capture images at specified waypoints or intervals. GPS data for each image is recorded for potential use in trajectory and pose determination. Additionally, datasets are collected in synthetic environments generated using tools like Unreal Engine. Virtual environments are designed with relevant features, and a virtual camera is programmed to follow specific trajectories, capturing images from various viewpoints within the environment.

Once images are collected, they are fed into 3D reconstruction software such as Meshroom or Colmap to determine trajectory and camera poses. If camera poses are known, they are inputted into the software, and the 3D reconstruction process is executed. However, if camera poses are unknown, the software employs Structure from Motion (SfM) techniques to determine them based on the images provided. The images can be segregated as Synthetic shown by fig. 3.3 and fig. 3.4. Real datasets can also be collected using setup in manual setting such as fig. 3.5 and fig. 3.6 as shown in fig. 3.7 and fig. 3.8

Figure 3.1: Terrestrial Robot - Copernicus

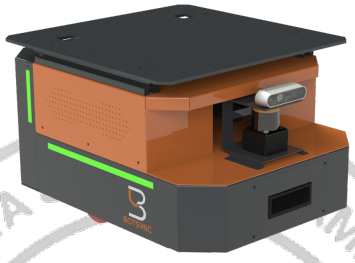


Figure 3.2: Terrestrial Robot - Volta

Figure 3.3: Synthetic Dataset 1

Figure 3.4: Synthetic Dataset 2

Figure 3.5: Aerial Dataset Collection

Figure 3.6: Manual Dataset collection setup

The next steps involve importing the 3D mesh and camera data obtained from the reconstruction software into Blender for visualization and further processing. Trajectory verification is performed to ensure the accuracy of imported camera trajectories, comparing them with the original capture trajectories and documenting any discrepancies or ambiguities. Scene scaling and transformation are then applied based on known camera poses, computing the scene scale using known distances between camera positions and aligning the 3D model accurately.

Finally, the finalized 3D reconstructed scene is utilized for various purposes, including research analysis and applications such as virtual simulations, architectural planning, or environmental analysis. This comprehensive process enables the creation of detailed and accurate digital representations of real-world scenes, facilitating a wide range of applications and insights.

Figure 3.7: Real Dataset 1

Figure 3.8: Real Dataset 2

3.3 Types of 3D Reconstruction

3D reconstruction can be categorized into two primary methods: sparse and dense 3D reconstruction. Sparse reconstruction focuses on identifying the structure of the scene using a limited number of key points, providing a foundational framework. In contrast, dense reconstruction aims to create a detailed and comprehensive 3D model by capturing the fine details and textures of the scene. Both methods are essential and often used sequentially in the 3D reconstruction pipeline to achieve high-quality and accurate 3D models.

3.3.1 Sparse 3D Scene Reconstruction

Sparse 3D scene reconstruction[47] is the initial phase in the 3D reconstruction process whose results are as shown in fig. 3.9 and fig. 3.10. It focuses on estimating the positions and orientations of the cameras and a sparse set of 3D points that represent the most distinctive features in the scene. This process is primarily driven by the Structure from Motion (SfM) technique, which involves the following steps:

The process of 3D reconstruction begins with several key stages that are crucial for creating an initial framework of the scene. The first step is feature detection, which involves identifying key points or features in the images. This is achieved using advanced algorithms like SIFT (Scale-Invariant Feature Transform) or ORB (Oriented FAST and Rotated BRIEF). These algorithms are designed to recognize distinctive patterns in the images that remain consistent across different views, such as corners, edges, or textured areas. By detecting these key points, the algorithms provide a set of features that can be reliably tracked and matched across multiple images.

Following feature detection is the feature matching stage. This step involves establishing correspondences between the detected features in different images. By comparing the key points identified in each image, the matching algorithm determines which points in one image correspond to points in another image. This correspondence is essential for understanding how different images relate to each other spatially, as it allows for the alignment of multiple viewpoints. Accurate feature matching ensures that the same physical points in the scene are consistently identified across different images, providing a basis for further geometric analysis.

Once feature matching is completed, the next step is camera pose estimation. This involves determining the position and orientation of each camera relative to the scene. By

Figure 3.9: Sparse Reconstruction

analyzing the matched features and their spatial relationships, algorithms can calculate how each camera was positioned and oriented when each image was captured. This step is crucial for understanding the spatial configuration of the cameras and the scene, as it provides the necessary information to reconstruct the scene in 3D. Accurate camera pose estimation ensures that the subsequent stages of reconstruction can build a coherent and accurate model of the scene.

The final stage in this initial phase is sparse point cloud generation. This involves triangulating the matched features to create a sparse point cloud representing the 3D structure of the scene. By using the positions and orientations of the cameras along with the matched features, algorithms can calculate the 3D coordinates of these features in the scene. The result is a sparse point cloud that provides a rough outline of the scene's structure, capturing the major geometric features. This sparse point cloud serves as the foundation for further refinement and detailing in subsequent stages of 3D reconstruction, where more points and finer details will be added to create a comprehensive and detailed model of the scene.

These steps—feature detection, feature matching, camera pose estimation, and sparse point cloud generation—are critical for establishing the initial framework of the scene in the 3D reconstruction process. They transform 2D image data into a basic 3D representation, setting the stage for further refinement and enhancement in the dense reconstruction phase.

The sparse reconstruction provides a rough outline of the scene and serves as the foundation for the subsequent dense reconstruction phase. It is computationally less

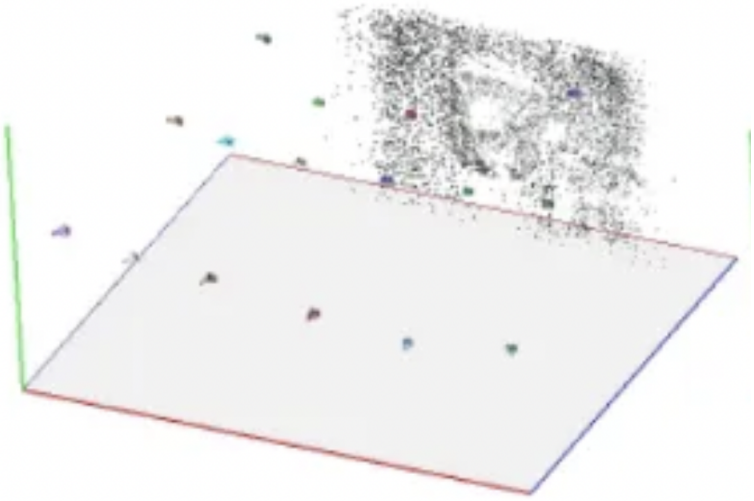


Figure 3.10: Sparse Reconstruction 2

intensive and faster, allowing for the initial alignment and verification of camera poses.

3.3.2 Dense 3D Scene Reconstruction

Dense 3D scene reconstruction[48] builds upon the sparse reconstruction by generating a comprehensive and detailed 3D model as shown in fig. 3.11. This phase involves the following steps:

The process of 3D reconstruction continues with several critical stages that transform the initial sparse point cloud into a detailed and realistic 3D model. Each stage plays a vital role in enhancing the accuracy and visual fidelity of the reconstructed scene.

The first step in creating a detailed 3D model is depth map estimation. For each image, a depth map is computed, providing the distance from the camera to the surfaces within the scene. This step is crucial as it transforms 2D image data into 3D information. Multi-View Stereo (MVS) techniques are commonly used for this purpose. MVS leverages multiple overlapping images to calculate depth, using the parallax effect where the apparent displacement of an object is observed from different viewpoints. By analyzing these disparities, MVS algorithms can accurately estimate the depth for each pixel in the images, resulting in depth maps that capture the geometric structure of the scene.

Once depth maps are obtained for all images, the next step is to merge these depth maps from multiple viewpoints to create a dense point cloud. This point cloud consists of a vast number of points, each representing a specific location in the 3D space of the scene. The dense point cloud provides a much more detailed and comprehensive

representation compared to the initial sparse point cloud, capturing intricate details of the scene's geometry. The process of merging depth maps involves aligning and integrating the information from different perspectives to ensure a cohesive and accurate 3D representation.

The dense point cloud is then converted into a mesh, which is a continuous surface composed of vertices, edges, and faces. Mesh generation involves connecting the points in the dense point cloud to form a network of polygons, typically triangles or quadrilaterals. This step transforms the discrete set of points into a structured and continuous surface that can be easily manipulated and analyzed. The resulting mesh represents the shape and structure of the scene with high fidelity, providing a foundation for further processing and visualization.

The final step in enhancing the visual realism of the 3D model is texturing. Texturing involves applying color information from the original images onto the mesh. By mapping the colors and textures from the 2D images onto the 3D surface, the model gains a more realistic and lifelike appearance. This process not only improves the visual quality of the 3D model but also helps in preserving the original details and characteristics of the scene. Texturing ensures that the 3D model accurately reflects the appearance of the real-world environment, making it suitable for various applications such as virtual simulations, visual effects, and architectural visualization.

Through these stages—depth map estimation, dense point cloud generation, mesh generation, and texturing—the initial sparse representation of the scene is transformed into a detailed and realistic 3D model. Each step builds upon the previous one, enhancing the geometric accuracy and visual fidelity of the reconstruction. The final 3D model serves as a rich and detailed representation of the real-world environment, enabling a wide range of applications and analyses.

Dense reconstruction provides a detailed and high-resolution model of the scene, capturing fine geometric details and textures. It is more computationally intensive and requires more processing time and resources compared to sparse reconstruction.

Sparse and dense 3D scene reconstruction[50] are integral components of the 3D reconstruction pipeline. Sparse reconstruction establishes the basic structure and camera positions, serving as the groundwork for the detailed dense reconstruction. Together, these processes enable the creation of accurate and detailed 3D models from 2D images,

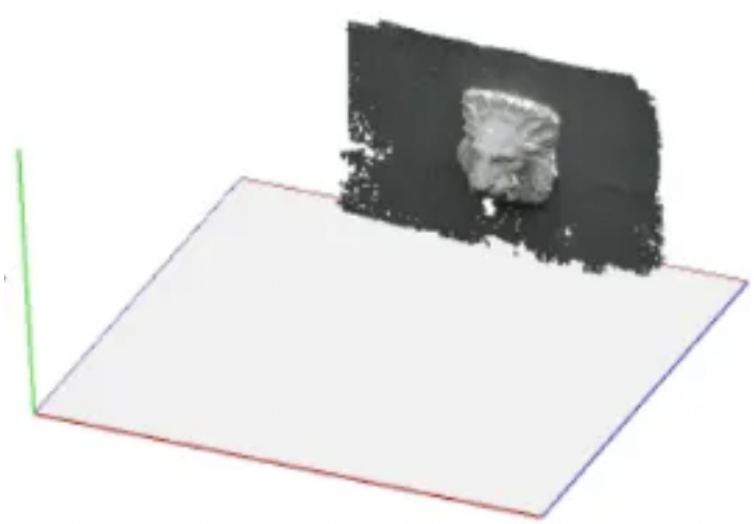


Figure 3.11: Dense Reconstruction

with applications in fields such as virtual reality, robotics, cultural heritage preservation, and more.

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Chapter 4

Reflections

CHAPTER 4

REFLECTIONS

3D reconstruction using real-life datasets collected from drones and terrestrial bots presents a cutting-edge method for creating detailed and accurate 3D models of environments. This technique involves capturing numerous images from different perspectives and then processing these images using sophisticated software to generate a cohesive 3D model. However, the process is fraught with challenges that can significantly impact the accuracy and quality of the final reconstruction. Among these challenges are issues related to image processing, such as inappropriate scaling, transformation, and rotation of the generated mesh, which can lead to distortions and inaccuracies in the 3D model.

4.1 Challenges Faced

Image Quality and Consistency

Variability in lighting, weather conditions, and camera quality can lead to inconsistent image data, making it difficult to achieve a seamless integration of images from drones and terrestrial bots.

Alignment and Registration

Ensuring that images from different sources align correctly is critical. Differences in vantage points and angles can cause difficulties in registering images accurately, leading to potential misalignment in the final model.

Scaling Issues

Accurately scaling the 3D model to real-world dimensions is a complex task. Inconsistent scaling factors from different datasets can result in disproportionate models, where certain parts are either exaggerated or minimized.

Transformation and Rotation Errors

Correctly transforming and rotating the meshes to fit together is challenging. Even slight errors in rotation or transformation can cause significant distortions, making the final 3D model unrealistic or unusable.

Data Integration

Integrating data from aerial drones with ground-based bots requires sophisticated algorithms to merge these datasets effectively. Differences in perspective and resolution between aerial and terrestrial images can complicate this integration.

Computational Load

Processing large datasets from high-resolution images requires substantial computational power and advanced algorithms, which can be resource-intensive and time-consuming.

Environmental Factors

External factors such as moving objects, changes in the environment between data captures, and obstructions can introduce noise and errors in the dataset, complicating the reconstruction process.

Software Limitations

The software used for 3D reconstruction may have limitations in handling diverse datasets and may not always provide the tools necessary for correcting scaling, transformation, and rotation issues effectively.

Addressing these challenges requires a combination of advanced image processing techniques, robust algorithms for data integration, and iterative refinement to ensure the accuracy and fidelity of the 3D models.

4.2 Learning Outcomes

The project provides a comprehensive understanding and practical experience in the field of 3D reconstruction from multi-source image datasets. The key learning outcomes include:

- **Proficiency in Image Collection Techniques**: Gain hands-on experience in collecting image data using terrestrial bots with SLAM, aerial drones, and synthetic environments generated by Unreal Engine. Understand the importance of trajectories and camera poses in capturing accurate datasets.
- **Knowledge of 3D Reconstruction Software**: Develop skills in using Meshroom and COLMAP software for 3D reconstruction. Learn to feed camera poses into

these programs for accurate scene reconstruction and understand the Structure-from-Motion (SfM) method for determining unknown camera poses.

- **Integration with Blender**: Learn how to import the generated mesh and corresponding camera data into Blender. Understand the process of verifying and correcting the trajectory of imported cameras to match the original paths followed during image capture.

- **Handling Scale, Translation, and Rotation**: Acquire the ability to compute the scale of the scene along with its translation and rotation when camera poses are known. Understand the impact of these factors on the accuracy of the final 3D model.

- **Application of 3D Models**: Explore the potential applications of the final constructed 3D scene in various fields such as virtual reality, simulation, urban planning, and environmental studies. Understand how to use these models for further research and practical applications.

- **Interdisciplinary Knowledge**: Gain insights into the interdisciplinary nature of the project, involving computer vision, robotics, photogrammetry, and 3D modeling. Understand how these fields converge to solve complex real-world problems.

- **Technical Documentation and Reporting**: Enhance skills in documenting technical processes and reporting findings. Learn to present complex workflows and results clearly and effectively for academic and professional purposes.

Overall, this project equips participants with a deep understanding of 3D reconstruction processes, practical skills in handling diverse datasets, and the ability to apply these skills to real-world scenarios.

Course Outcome and Project Outcomes Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	H	H	H	H	M	M	L	L	M	H	H	M	L	L	H	H
CO2	H	H	H	M	H	L	L	L	M	M	M	H	L	L	H	M
CO3	H	H	M	M	M	H	H	L	H	M	M	L	L	L	M	H
CO4	L	H	L	H	H	H	H	H	M	L	L	M	L	L	H	M

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